

Multilingual Low-resource Grapheme-to-Phoneme Conversion

Aosimanjiang Aihaiti, Yanbo Luo and Yi Wu, supervised by Shu Okabe

Introduction

Grapheme-to-Phoneme Conversion (G2P): A computational process that maps sequences of **graphemes** (the basic written units) to sequences of **phonemes** (the basic sound units, represented with International Phonetic Alphabet), core component of text-to-speech (TTS) and automatic speech recognition (ASR) systems

Low-resource: Simulates a low-resource setting, each language consists of **1,000** words

Data Processing

Dataset Origin

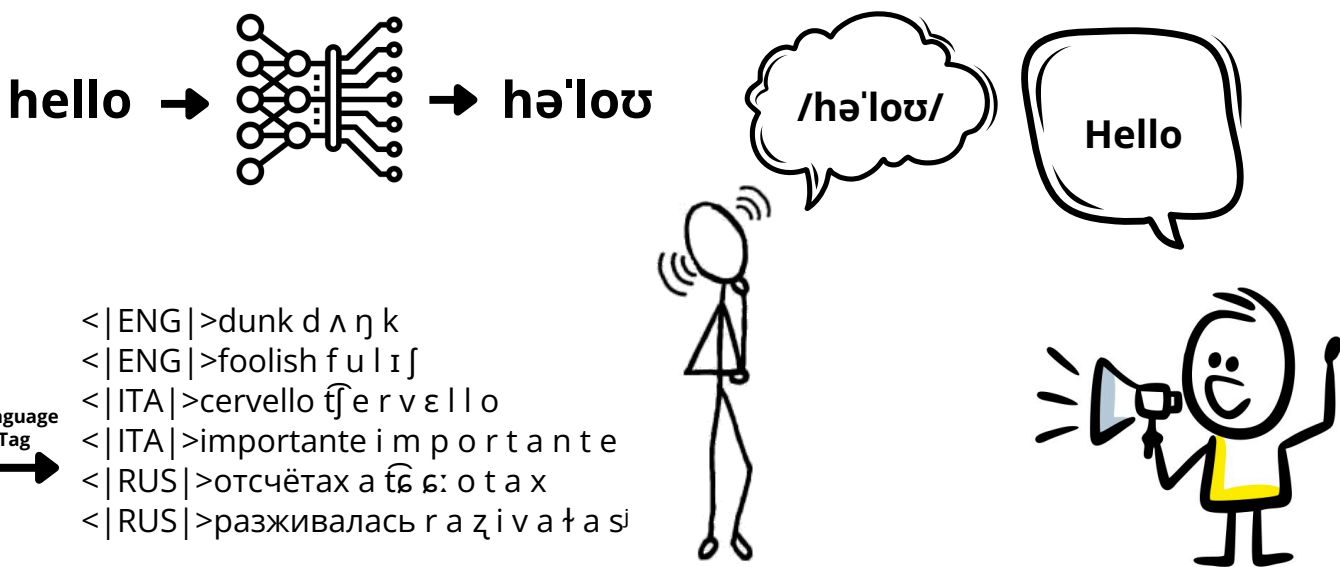
- SIGMORPHON Shared Task: Grapheme-to-Phoneme (year 2020, 2021, 2022, 2024)
- WikiPron's "Big Scrape" Dataset, scraped from Wiktionary
- Oversampled for languages with data fewer than 1000 words

Family/Script	Branch	Latin	Cyrillic	Arabic	Greek	Armenian	Hangul
Indo-European	Germanic	English					
	Italic	Italian					
	Balto-Slavic	Polish	Russian				
	Hellenic				Greek		
	Armenian					Armenian	
Indo-Iranian				Persian			
Austronesian		Indonesian					
Afro-Asiatic		Maltese		Arabic			
Turkic				Uyghur			
Koreanic							Korean

Mono to Multilingual Dataset Pipeline

dunk dʌŋk + cervello tʃɛrvɛllo + отсчётах а tɕɕɔ:təx
foolish fu:lɪʃ + importante i:m.pɔ:rtante + разживалась rəʒi.vəˈlʌ.sɪ

Language Tag



Models

- Transformer:** A standard attention-based sequence-2-sequence transformer model
- Transducer:** A majority-vote ensemble of neural transducers trained with imitation learning and adopted on G2P task
- Attentive LSTM:** Long-short Term Memory neural network (LSTM) with soft attention mechanism

Evaluation Criteria

- Word Error Rate (WER):** (Number of Incorrect Predictions / Total Number of Words) × 100%

Results

Comparison between models

Performance of different Models in monolingual Setting (Average WER)

Neural Transducer	28.98%
Attentive LSTM	32.06%
Transformer	35.23%

Performance of different Models in multilingual Setting

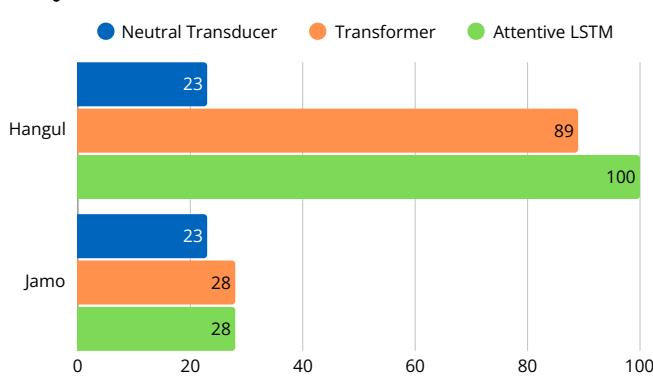
Models	#Multi > Mono	#Multi < Mono	Average Imprv
Transducer	15	3	-5.11
Transformer	6	12	2.44
Attentive LSTM	6	12	1.33

Best vs. Worst Performers

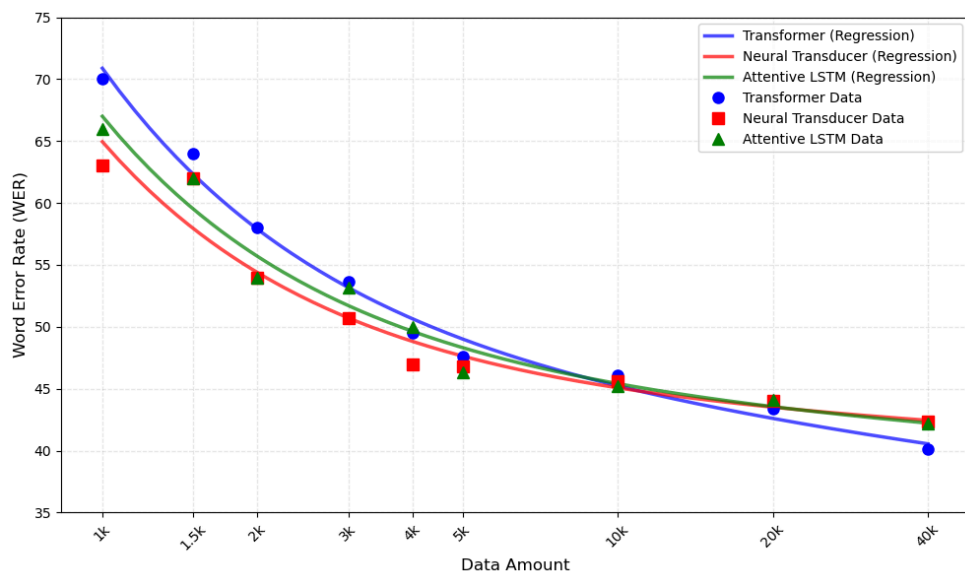
Languages with lowest WER:
Georgian & Uyghur (Both 1.25%)

Languages with highest WER:
Korean (73.25%)

Optimization by decomposing characters



All models WER Performance Across Varying English Dataset Sizes

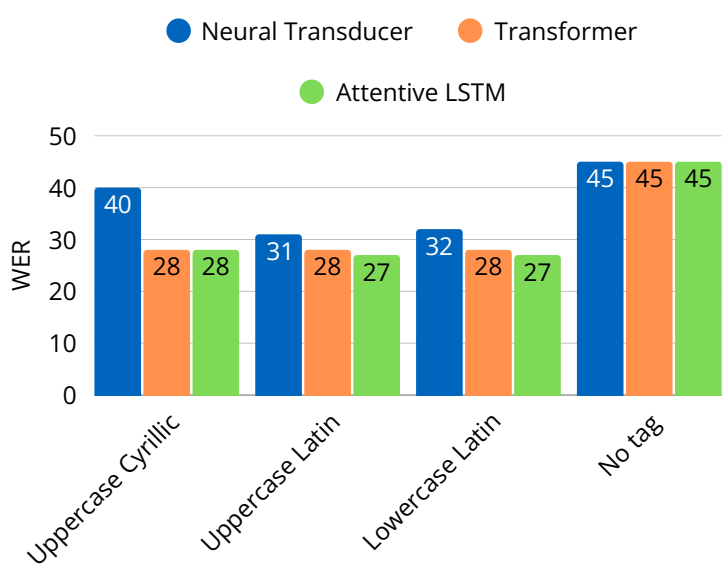


Multilingual

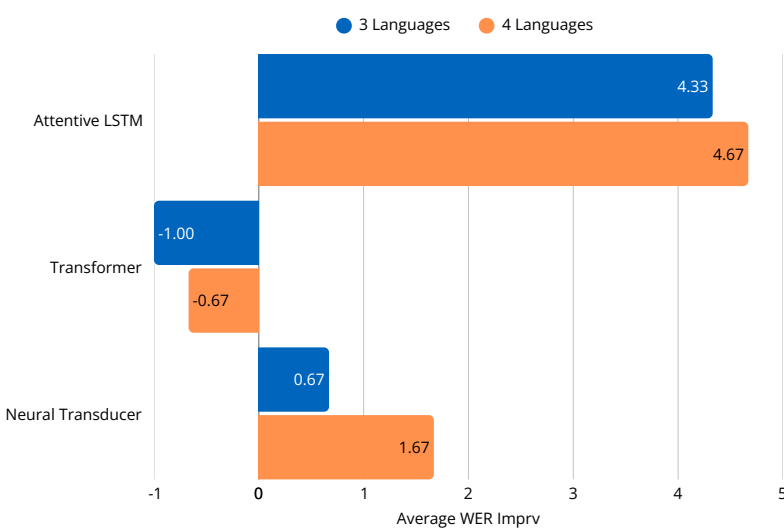
To explore the multilingual capabilities of our G2P model, we conducted more experiments:

- Analyse effect of different script-based language tags
- Assess the effect of varying the number of languages in the multilingual training set
- Measure performance improvement over monolingual models
- Investigate cultural Influence of languages cross language family
- Compare model performance on language in native script vs. Latin transliteration

1. Tag Script Exploration



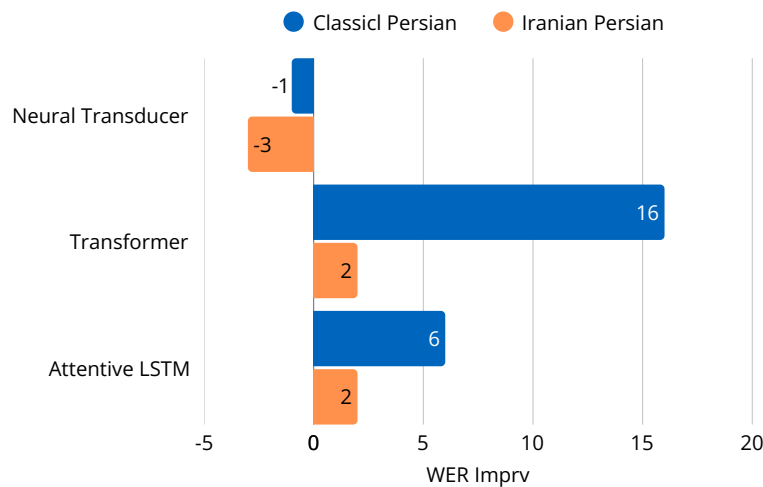
2. Dataset Size Impact In Multilingual Setting



3. Most improved Datasets using Transformer

Language Family	Script	Language	Imprv
Indo-European (Germanic)	Latin	German	6
		Swedish	9
		Dutch	9
Average			8
Afro-Asiatic	Arabic	Arabic	5
Indo-European		Classical Persian	16
Turkic		Uyghur	0
Average			7
Indo-European	Arabic	Classical Persian	18
	Cyrillic	Russian	0
	Latin	Italian	-1
Average			5.67

4. Persian and Uyghur in the same Dataset



5. Transliteration Example

Arabic	Uroman	Epitrans	IPA
تشيلي	tshyly	tʃi:li:	tʃi:li:j
توفيق	twfyq	tu:fi:q	tawfi:q

Transliterated Arabic-Script Languages Result

